Ex:- 3b2
Date:
23/7/26

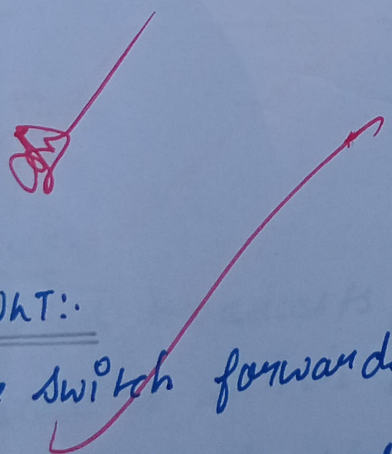
SWITCH

AIM:

To study the behavior of a switch by observing how it forwards data packets using cisco packet tracer.

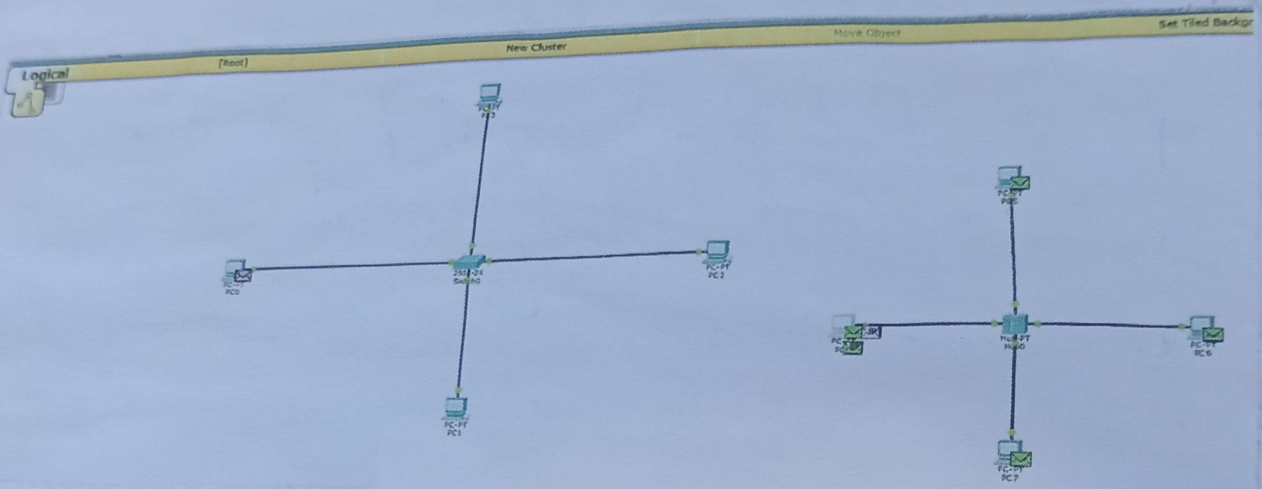
Steps:

- Open Cisco packet Tracer.
- Drag and place 1 switch and 4 generic PCs into the workspace.
- Connect all PCs to the switch using copper straight-through cables.
- Configure IP addresses for all PCs in the same networks.
- Select Add Simple PDU from the common Tools bar.
- Send a packet from one PC (source) to another PC (destination).
- Switch to simulation mode and observe how the switch forwards the packet.



RESULT:

The switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary networks traffic and improves communication efficiency


Logical
[Root]
New Cluster
Move Object
Set Tiled Backgr

Ex: 3b3
Date: 23/7/26

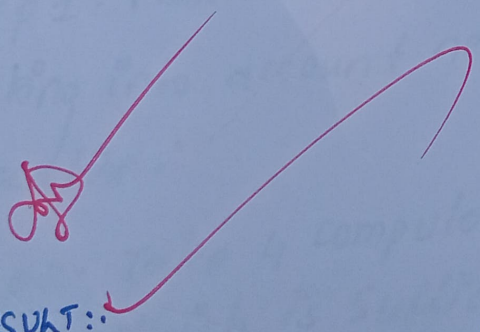
HUB and SWITCH (comparision)

AIM:

To compare the packet forwarding behavior of a Hub and Switch using Cisco packet Tracer.

STEPS:

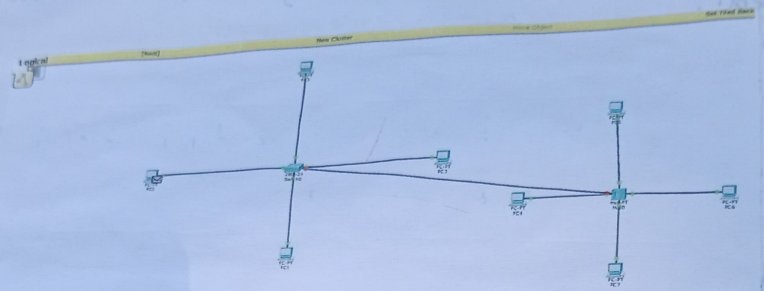
- Create one network using a Hub and another using a Switch, each connected to 4 PCs.
- Configure IP addresses for all PCs in both networks.
- Send a simple PDU between two PCs in each networks.
- Observe packet transmission in simulation mode.
- Compare how the hub and switch forward packets.



RESULT:

The hub broadcasts packets to all connected devices, whereas the switch forwards packets only to the intended destination. Therefore, the switch provides better performance, reduced network traffic, and improved efficiency compared to the hub.

Ex: 3b3
Date:
23/7/24



PC0

IP Configuration

IP Configuration

DHCP Static

IP Address 10.1.1.1

Subnet Mask 255.0.0.0

Default Gateway

DNS Server

PC1

IP Configuration

IP Configuration

DHCP Static

IP Address 10.1.1.2

Subnet Mask 255.0.0.0

Default Gateway

DNS Server

EX:-4
Date:-
2/8/26

LAN using a switch and Ethernet cables.

AIM:-

Setup and configure a LAN (Local area network) using a switch & Ethernet cables in your lab.

What is a LAN?

A local Area network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers, and internet information between users, such as computers, printers, servers, and switches. A local area network (LAN) switch serves as the primary connecting, device managing and directing communications within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

Steps to set up a LAN:-

Step 1. Plan and Design an appropriate network topology taking into account network requirements and equipment location.

Step 2. Take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet cables.

Step 3: Connect the computers to network switch via an Ethernet cables, which is as simple as plugging one end of the Ethernet cables into the computer and the other end into the network switch.